

Inductive Venn–Abers and related regressors

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Abstract

Venn–Abers predictors are probabilistic predictors that enjoy appealing properties of validity, but their major limitation is that they are applicable only to the case of binary classification, with a recent extension to bounded regression. We generalize them to the case of unbounded regression, which requires adding an element of conformal prediction. In our simulation and empirical studies we investigate the predictive efficiency of point regressors derived from Venn–Abers regressors and argue that they somewhat improve the predictive efficiency of standard regressors for larger training sets.

Keywords: Perfect calibration, inductive Venn–Abers regressor, cross Venn–Abers regressor.

1. Introduction

This paper explores the problem of regression estimation as presented in, e.g., [Vapnik \(1998, Sect. 1.4\)](#). In the standard setting of statistical learning, where we observe an IID sequence of random pairs (X, Y) consisting of objects X and their labels $Y \in \mathbb{R}$, the (ideal) *regression estimator* (of Y given X) maps each object x to the expected value $\mathbb{E}(Y \mid X = x)$ of its label Y conditional on observing x . Vapnik lists regression estimation as one of three basic statistical problems ([Vapnik, 1998, Sect. 1.2](#)). We develop ideas of conformal prediction ([Vovk et al., 2022](#); [Angelopoulos et al., 2026](#)) and introduce an algorithm for regression estimation that satisfies a natural distribution-free notion of validity.

We are interested in the distribution-free setting, when nothing is known about the distribution generating one pair (X, Y) and we are only given a training sequence of labelled objects and an unlabelled test object. In typical cases we cannot hope to find the true regression estimate $\mathbb{E}(Y \mid X = x)$ (even when it is well-defined), and so we lower the bar in three respects in our definition of a valid regression estimator. First, we allow estimators of the form $\mathbb{E}(Y \mid \mathcal{F})$ for some σ -algebra $\mathcal{F}$; we will express this by saying that our regression estimator is perfectly calibrated. Ideally, $\mathcal{F}$ should be close to the σ -algebra generated by the training set and test object. Second, we allow our algorithms to output intervals (ideally short “imprecise regression estimates”) containing $\mathbb{E}(Y \mid \mathcal{F})$. And third, we replace $\mathbb{E}(Y \mid \mathcal{F})$ by $\mathbb{E}(Y' \mid \mathcal{F})$, where Y' is a “regularized” version of Y such that $Y' = Y$ with high probability. (This is the element of conformal prediction that we mentioned in the abstract; it is required in the absence of bounds on the test label.)

As usual, we consider two principal requirements for our algorithms, validity (technically, perfect calibration) and efficiency. Validity (in this technical sense) will be guaranteed for

our algorithms, but to achieve efficiency we will use existing regression algorithms that we believe to be efficient, although perhaps miscalibrated. We will develop methods for improving their calibration. These methods will be adaptations of the methods used in Vovk et al. (2015) (and presented in a much greater detail in Vovk et al. 2022, Chap. 6) in the context of binary classification, with bounded regression considered earlier by van der Laan and Alaa (2024).

The problem of regression estimation considered in this paper is very different from the regression problems in conformal prediction, where the task is to output a prediction region (typically a prediction interval) for a test label with a pre-specified coverage probability. In regression estimation the task is to cover the expected label $\mathbb{E}(Y \mid \mathcal{F})$ rather than the label Y itself. This will allow us to produce much shorter intervals (especially that we replace Y by Y' that is not guaranteed to coincide with Y).

The main goal of the computational experiments reported in this paper is to demonstrate that application of our methods leads to an improvement in the performance of standard point regressors (we consider those implemented in `scikit-learn`). The improvement is not as significant as we had hoped and disappears for smaller datasets.

2. Comparisons with Literature

Results of this paper are not directly comparable with the existing literature because of a somewhat unusual notion of validity that we use. But there are several related approaches.

A strand of research that is closely related to what we do in this paper is conformal regression, already mentioned in the previous section. Important advances in conformal regression include Romano et al. (2019), offering very flexible methods based on quantile regression, and Gibbs et al. (2025), establishing conditional validity results. See, e.g., Angelopoulos et al. (2026) for a recent review of conformal prediction in general and conformal regression in particular.

The goal of conformal regression is to produce provably valid, under the assumption of IID data, prediction intervals for the label of a test object. Since validity here means a guaranteed coverage probability, this is a much more ambitious problem than regression estimation dealt with in this paper, as mentioned earlier.

An even more closely related, but still very different, problem is that of covering the conditional expectation $\mathbb{E}(Y \mid X = x)$ for a given test object x . Interestingly, for a nonatomic distribution of objects, this problem is as difficult as predicting the label Y itself (Angelopoulos et al., 2026, Theorem 11.3).

Conformal predictive distributions (Vovk et al., 2022, Chap. 7) are different from conformal regression in that, instead of prediction intervals, they output full predictive distributions for future labels (assumed to be real-valued, as in conformal regression). To achieve validity these predictive distributions should be imprecise in a certain sense. Our current task is easier in that we only aim to cover $\mathbb{E}(Y' \mid \mathcal{F})$, but we will achieve another property of validity, namely perfect calibration instead of calibration in probability (the two notions of calibration are not comparable in general). See Allen et al. (2025) for a recent analysis of achievable properties of validity for conformal predictive distributions.

The property of validity used in this paper is inherited from Venn prediction, which is, however, typically applied to produce imprecise probability forecasts in classification prob-

lems. Venn prediction (including Venn–Abers prediction) is reviewed in Vovk et al. (2022, Chap. 6) and Venn–Abers prediction is reviewed in Angelopoulos et al. (2026, Sect. 12.4).

Venn–Abers predictors were extended to the case of bounded regression by van der Laan and Alaa (2024), as mentioned earlier. Our bounded Venn–Abers regressor introduced in the next section is just an inductive version of Algorithm 1 in van der Laan and Alaa (2024). As the next step, van der Laan and Alaa develop an algorithm (van der Laan and Alaa, 2024, Algorithm 2) for “self-calibrating conformal prediction” based on their Algorithm 1 (so that bounded Venn–Abers regression is only a small part of van der Laan and Alaa 2024). Finally, van der Laan and Alaa (2025) extend these results to general loss functions interpreting the usual regression setting as the case of squared error loss.

A key step in our main algorithm is replacing the true labels Y by their regularized versions Y' , as discussed in the previous section. We need it to avoid the vacuous regression interval $(-\infty, \infty)$ (see the end of Sect. 3). Assumptions made by van der Laan and Alaa (2024) and van der Laan and Alaa (2025) allow them to bypass this step, but we avoid making any assumptions on the data-generating distribution apart from the observations being IID.

In general, the properties of validity of our procedures are sufficiently different from the properties of validity considered in literature to make direct comparison between them in empirical or simulation studies difficult or impossible. Therefore, in our experimental section (Sect. 8) we concentrate on evaluating the predictive efficiency of point regressors derived from imprecise regressors implementing our methods; namely, we compare the predictive performance of those point regressors with that of the base algorithms.

3. Inductive Venn–Abers Regressors

In this section we ignore the computational complexity of our methods (it will be one of the topics of Sect. 5). Fix a measurable space $\mathbf{X}$ (the *object space*) and set $\mathbf{Z} := \mathbf{X} \times \mathbb{R}$ (this is the *example space*). Each example $z = (x, y) \in \mathbf{Z}$ consists of an object $x \in \mathbf{X}$ and its real-valued label y .

We consider two settings (leading to formally different, albeit similar, prediction algorithms). In the setting of *bounded regression* we are given a finite interval $[C_*, C^*] \subseteq \mathbb{R}$ guaranteed to contain all labels, training and test. Otherwise, we have *unbounded regression*. In the former case, the algorithm and its property of validity are simpler, and we consider them separately even though we are not particularly interested in them per se as we would like to avoid any assumptions apart from IID observations.

Notice that an algorithm for unbounded regression can also be applied in the bounded setting, and it may well be preferable if the bounds C_* and C^* are loose. This is another reason why our main interest is in unbounded regression.

3.1. Bounded regression

Fix any regression algorithm (such as a neural net) producing point predictions as the *base algorithm*. We are given a training set consisting of $l > k$ examples (where $k \in \mathbb{N} := \{1, 2, \dots\}$ is typically large and $l - k \in \mathbb{N}$ is also large) and a test object $x \in \mathbf{X}$; k is a parameter of our algorithm.

In bounded regression, we are given an interval $[C_*, C^*]$ guaranteed to contain all labels. The corresponding *bounded inductive Venn–Abers regressor* (bounded IVAR) produces the *regression interval*

$$[\hat{y}_*, \hat{y}^*] := [f_*(r), f^*(r)] \quad (1)$$

for the test label, where f^* , f_* , and r are defined as follows:

1. Randomly split the training set of size $l > k$ into two parts, a *proper training set* of size $l - k$ and a *calibration set* $z_1, \dots, z_k$ of size k ; for each $i \in \{1, \dots, k\}$, $z_i = (x_i, y_i)$ consists of an object x_i and its label y_i .
2. Train the regression algorithm on the proper training set obtaining a *prediction rule* $R : \mathbf{X} \rightarrow \mathbb{R}$ (a measurable function) mapping objects to their predicted labels.
3. Find the prediction $r_i := R(x_i)$ (*base prediction*) for the calibration object x_i , $i = 1, \dots, k$, and the base prediction $r := R(x)$ for the test object x .
4. Fit isotonic regression to $(r_1, y_1), \dots, (r_k, y_k), (r, C^*)$ obtaining an isotonic calibrator f^* .
5. Set $\hat{y}^* := f^*(r)$.
6. Fit isotonic regression to $(r_1, y_1), \dots, (r_k, y_k), (r, C_*)$ obtaining an isotonic calibrator f_* .
7. Set $\hat{y}_* := f_*(r)$.

The bounded IVAR is a simple modification of the inductive Venn–Abers predictor as defined in [Vovk et al. \(2015\)](#). First, we relax the condition $y_i \in \{0, 1\}$ of binary labels by allowing the labels to take intermediate values, $y_i \in [0, 1]$, and then we scale the interval $[0, 1]$ to arbitrary $[C_*, C^*]$. As mentioned in the previous section, non-inductive Venn–Abers regressors were introduced by [van der Laan and Alaa \(2024\)](#).

3.2. Unbounded regression

We again fix a regression algorithm and are given a training set of size $l > k$. Another parameter of our algorithm is $m \in \{1, \dots, \lfloor (k - 1)/2 \rfloor\}$; we are interested in a small m , such as $m = 1$. The corresponding *inductive Venn–Abers regressor* (IVAR) produces the following regression interval of the form (1) for the test label:

1. Randomly split the training set of size $l > k$ into two parts, a *proper training set* of size $l - k$ and a *calibration set* $z_1, \dots, z_k$ of size k , as before.
2. Train the regression algorithm on the proper training set obtaining a prediction rule $R : \mathbf{X} \rightarrow \mathbb{R}$.
3. Find the base predictions $r_i := R(x_i)$, $i = 1, \dots, k$, and $r := R(x)$.

4. Replace the m smallest calibration labels y_i by the $(m+1)$ th smallest calibration label y_* and replace the $m-1$ largest calibration labels y_i by the m th largest calibration label y^* . (Notice the asymmetry in the definitions of y_* and y^* .) In other words, let the new calibration labels be

$$y'_i := \begin{cases} y_* & \text{if } y_i < y_* \\ y^* & \text{if } y_i > y^* \\ y_i & \text{otherwise.} \end{cases} \quad (2)$$

(Notice that the recipe is still unambiguous when there are ties among y_i .)

5. Fit isotonic regression to $(r_1, y'_1), \dots, (r_k, y'_k), (r, y^*)$ obtaining an isotonic calibrator f^* .
6. Set $\hat{y}^* := f^*(r)$.
7. Replace the m largest calibration labels y_i by the $(m+1)$ th largest calibration label y^* and replace the $m-1$ smallest calibration labels y_i by the m th smallest calibration label y_* . (Notice that these y_* and y^* are different from those in step 4.) In other words, let the new calibration labels be defined as (2) for the new y_* and y^* .
8. Fit isotonic regression to $(r_1, y'_1), \dots, (r_k, y'_k), (r, y_*)$ obtaining an isotonic calibrator f_* .
9. Set $\hat{y}_* := f_*(r)$.

When fitting isotonic regression in steps 5 and 8 (and the analogous steps, 4 and 6, in the bounded IVAR), we always use the standard pool-adjacent-violators algorithm (PAVA; see, e.g., Barlow et al. 1972, Sect. 1.2). Notice that the new steps 4 and 7 in the IVAR as compared with the bounded IVAR are really needed: if we just set $y^* := \infty$ and $y_* := -\infty$, the PAVA will produce $(-\infty, \infty)$ as the regression interval.

4. Validity

We often write $Z_1, \dots, Z_k$, where $Z_i = (X_i, Y_i)$, for the calibration examples and (X, Y) for the test example, in order to emphasize that they are considered as random elements.

First we state a simpler property of validity, the one for bounded regression (including binary classification as special case). So we assume that the labels take values in $[C_*, C^*]$. Let us say that a random variable S is *perfectly calibrated* as regression estimate of Y if $S = \mathbb{E}(Y \mid S)$ a.s. (In our terminology we follow, e.g., Vovk et al. 2022, Sect. 6.2.1, Angelopoulos et al. 2026, Chap. 12, van der Laan and Alaa 2024, and van der Laan and Alaa 2025; another popular term is “auto-calibration”: see, e.g., Krüger and Ziegel 2021, Definition 3.1.)

Remark 1 An equivalent definition of perfect calibration is that a random variable S is said to be perfectly calibrated as regression estimate of Y if $S = \mathbb{E}(Y \mid \mathcal{F})$ a.s. for some σ -algebra $\mathcal{F}$ (in which case we may also say that S is *ideal* relative to $\mathcal{F}$).

A *selector* for an IVAR is a random variable that always belongs to the regression interval output by the IVAR.

Theorem 2 *For any bounded IVAR, there is a selector S that is perfectly calibrated for the test label, i.e., $\mathbb{E}(Y \mid S) = S$ a.s.*

Our interpretation of Theorem 2 is that the regression interval produced by the bounded IVAR is approximately calibrated provided it is narrow enough.

Now we consider the case of unbounded regression and continue to assume $2m < k$. A *selector* is defined as before, and the *Winsorized* test label is defined by

$$Y' := \begin{cases} Y_{(m)} & \text{if } Y < Y_{(m)} \\ Y_{(k-m+1)} & \text{if } Y > Y_{(k-m+1)} \\ Y & \text{otherwise,} \end{cases} \quad (3)$$

where $Y_{(1)} \leq \dots \leq Y_{(k)}$ is the sequence $Y_1, \dots, Y_k$ of calibration labels sorted in the ascending order. We can regard (3) as a more feasible version of Y corrected for the possibility of Y being an outlier; we make it easier to predict by restricting it to be in the range of the calibration labels (perhaps with a cushion).

Remark 3 See Dixon (1960, Sect. 1) and Tukey (1962, Sect. 14) for the original definition of Winsorization, which we use in this paper. This definition is sometimes modified, as in, e.g., Wilcox (2012, Sect. 2.2.2). While the original definition is about moderating a dataset, the modified definition is about moderating the population.

Theorem 4 *We have $Y = Y'$ with probability at least $1 - \frac{2m}{k+1}$. For any IVAR, there is a selector S that is perfectly calibrated for the Winsorized test label Y' , i.e., $\mathbb{E}(Y' \mid S) = S$ a.s.*

The obvious first statement of Theorem 4 says that the Winsorized label Y' is the same as the original label Y with high probability assuming $m \ll k$. The second statement asserts the validity of the IVAR as regression function for the Winsorized label. See Appendix A for the proofs. They will make clear how the symmetry in the definition (3) of the Winsorized test label (involving the m th smallest and m th largest calibration label) leads to the asymmetry in the definition of IVARs in Sect. 3.2.

An alternative parametrization of the IVAR is in terms of $\epsilon := 2m/(k+1)$; we then have $Y = Y'$ with probability at least $1 - \epsilon$. If a given $\epsilon \in [2/(k+1), 1)$ is not of the form $2m/(k+1)$ for an integer m , we decrease it as little as possible so that it takes this form.

5. Algorithm

The procedure described in Sect. 3.2 can be implemented very efficiently. In our description of the algorithm, we will just refer to Vovk et al. (2022, Chap. 6, especially Sect. 6.5.3), which in turn follows Vovk et al. (2015). We will discuss computing f^* and f_* , which are denoted by f^1 and f^0 , respectively, in Vovk et al. (2022, Sect. 6.5.3). Remember that we moderate the original calibration labels y_i to their less extreme versions y'_i by (2); after that,

we forget the original labels and drop the primes. But it should always be remembered that y_i stand for the moderated labels.

First we see how to compute F^1 , which determines $f^1 = f^*$. We are given the base predictions r_i and the moderated labels $y_1, \dots, y_k$. Define k' , $r'_1, \dots, r'_{k'}$, $w_1, \dots, w_{k'}$, and $y'_1, \dots, y'_{k'}$ as in Vovk et al. (2022, Sect. 6.5.3) (where base predictions were called scores and were denoted by s_i). The CSD consisting of the points P_i , $i \in \{0, \dots, k'\}$, is defined as before, but now it is extended by adding the point $P_{-1} := (-1, -y^*)$; this corresponds to adding the test example to the calibration set assuming that the base prediction for it is smaller than the base prediction for any calibration example while its label is y^* (which is a most unusual combination). Algorithm 6.3 in Vovk et al. (2022, Sect. 6.5.3) will compute the corners of the resulting CSD $P_{-1}, \dots, P_{k'}$ and Algorithm 6.4 will do the rest.

Computing F^0 , which determines $f^0 = f_*$, is analogous. Now the CSD consisting of the points P_i , $i \in \{0, \dots, k'\}$, is extended by adding the point $P_{k'+1} := P_{k'} + (1, y_*)$; this corresponds to adding the test example to the calibration set assuming that its base prediction is larger than any calibration base prediction while its label is y_* (another most unusual combination). Algorithm 6.5 in Vovk et al. (2022, Sect. 6.5.3) will compute the corners of the resulting CSD $P_0, \dots, P_{k'+1}$ and Algorithm 6.6 will do the rest.

Algorithm 6.7 in Vovk et al. (2022, Sect. 6.5.3) shows how to arrange the calibration set into a convenient binary search tree, and Algorithm 6.8 shows how to use this tree for computationally efficient prediction. The computational complexity of these procedures is summarized in the following theorem (which ignores the complexity of training the base regressor and computing base predictions).

Theorem 5 *The computation time of our prediction algorithm is $O(k \log k)$ for preprocessing ($O(k)$ apart from sorting the calibration set). Once preprocessing is completed, processing each test object can be done in time $O(\log k)$.*

6. Merging a Regression Interval into a Single Value

The IVAR introduced in Sect. 3 achieves our goal of producing provably valid regression intervals that have a potential to be predictively efficient for efficient base algorithms. However, in order to be able to compare the predictive efficiency of our methods with traditional regression algorithms, in this and following section we will define natural modifications of the IVAR that produce point predictions. In this section we see how to replace the regression intervals output by IVARs by point predictions, and in the following section we will see how we can combine several IVARs to achieve both predictive and computational efficiency.

Given $y_* < \hat{y}_* < \hat{y}^* < y^*$, let us see how to replace the regression interval $[\hat{y}_*, \hat{y}^*]$ with a single regression value $\hat{y}$. Following the minimax approach of Vovk et al. (2022, Sect. 6.4.3), we need to solve the equation

$$(\hat{y} - y_*)^2 - (\hat{y}_* - y_*)^2 = (y^* - \hat{y})^2 - (y^* - \hat{y}^*)^2. \quad (4)$$

It is clear that there is a unique solution $\hat{y}$ in the interval $[\hat{y}_*, \hat{y}^*]$, since, over that interval, the left-hand side of (4) increases in $\hat{y}$ from 0 to a positive number, while the right-hand side decreases from a positive number to 0.

After simplification (4) becomes a linear equation in $\hat{y}$ with solution

$$\hat{y} = \frac{\hat{y}_*^2 - \hat{y}^{*2} + 2\hat{y}^*y_* - 2\hat{y}_*y_*}{2(y^* - y_*)}, \quad (5)$$

and there is only one solution (both in the interval $[\hat{y}_*, \hat{y}^*]$ and overall).

We can also solve (4) approximately obtaining a much more intuitive expression. Rewriting (4) as

$$2(\hat{y} - \hat{y}_*)(\hat{y}_* - y_*) + (\hat{y} - \hat{y}_*)^2 = 2(\hat{y}^* - \hat{y})(y^* - \hat{y}^*) + (\hat{y}^* - \hat{y})^2,$$

assuming $\hat{y}_* \approx \hat{y}^*$, and ignoring the quadratic terms, we obtain the approximate equation

$$(\hat{y} - \hat{y}_*)(\hat{y}_* - y_*) = (\hat{y}^* - \hat{y})(y^* - \hat{y}^*).$$

Regrouping its terms as

$$\hat{y}(\hat{y}_* - y_* + y^* - \hat{y}^*) = \hat{y}_*(\hat{y}_* - y_*) + \hat{y}^*(y^* - \hat{y}^*),$$

we can write its solution as the weighted average

$$\hat{y} = \frac{\hat{y}_* - y_*}{\hat{y}_* - y_* + y^* - \hat{y}^*} \hat{y}_* + \frac{y^* - \hat{y}^*}{\hat{y}_* - y_* + y^* - \hat{y}^*} \hat{y}^* \quad (6)$$

of $\hat{y}_*$ and $\hat{y}^*$.

7. Cross Venn–Abers Regressors

We define *cross Venn–Abers regressors* (CVARs) as in Vovk et al. (2022, Sect. 6.4.4) except that the regression interval coming from each fold is replaced by the regression estimate computed as described in Sect. 6, either as in (5) or as in (6) (in our experiments in the next section we use the latter). Namely, we divide the training set into K folds of approximately equal size, and use each fold in turn as the calibration set while the union of the remaining folds is used as the proper training set. The overall regression estimate is found as the arithmetic mean of the K regression estimates obtained by merging the regression intervals coming from the K IVARs corresponding to the K folds. In our experiments we set the parameter K to 10.

8. Experimental Results

As discussed at the end of Sect. 2, in our experimental studies we concentrate on our point regressors, namely the CVARs as defined in the previous section. These point regressors no longer satisfy any formal properties of validity, but the hope is that the validity properties of IVARs will show in superior performance of CVARs as compared with standard point regressors and measured in a traditional way.

We evaluate our approach on a suite of controlled synthetic regression benchmarks designed to isolate different statistical challenges. We generate datasets with $n = 10000$ examples under the following generative scenarios. Each dataset consists of n examples

whose objects are vectors of $d = 10$ features. As before, the objects are denoted by x_i and the corresponding labels by y_i , $i = 1, \dots, n$. Across all datasets, the level of noise in the generated labels is parametrized by $\sigma \in \{1, 3\}$. The results in this section represent the more challenging random noise level $\sigma = 3$, and results for $\sigma = 1$, as well as for $n = 1000$, are included in Appendix B. Unless stated otherwise, all random draws are independent.

Bounded Logistic dataset Each object is generated as $x_i \sim \mathcal{N}(0, I_{10})$, the weight vector as $w \sim \mathcal{N}(0, I_{10})$, and the Gaussian noise variables as $\xi_i \sim \mathcal{N}(0, \sigma^2)$. (We parametrize the Gaussian distribution $\mathcal{N}$ by its mean and variance, or its covariance matrix in the multidimensional case.) The labels are then generated using the sigmoid function:

$$y_i := \frac{10}{1 + e^{-w^\top x_i}} + \xi_i.$$

Notice that while the conditional expectation of y_i is bounded (belongs to $(0, 10)$), the labels are not bounded because of the Gaussian noise. Since y_i are “almost bounded” (the Gaussian distribution having thin tails), this is the most benign case, close to the setting of Sect. 3.1, and our methods work best for it.

Linear Gaussian dataset We generate $x_i \sim \mathcal{N}(0, I_{10})$, $w \sim \mathcal{N}(0, I_{10})$, and $\xi_i \sim \mathcal{N}(0, \sigma^2)$ as before. The labels are then defined by

$$y_i := w^\top x_i + \xi_i.$$

Nonlinear dataset Again $x_i \sim \mathcal{N}(0, I_{10})$ and $w \sim \mathcal{N}(0, I_{10})$. The label of x_i combines a linear contribution with smooth nonlinear components:

$$y_i := w^\top x_i + 2 \sin(x_{i,1}) + \frac{1}{2} x_{i,2}^2 - \cos(2x_{i,3}) + \xi_i,$$

where $\xi_i \sim \mathcal{N}(0, \sigma^2)$ and $x_{i,j}$ denotes the j th feature of x_i .

Heteroscedastic noise dataset We again draw $x_i \sim \mathcal{N}(0, I_{10})$ and $w \sim \mathcal{N}(0, I_{10})$ independently. However, the observation-noise scale now depends on the object. Specifically,

$$y_i := w^\top x_i + \xi_i, \quad \xi_i \sim \mathcal{N}(0, \sigma_i^2),$$

where

$$\sigma_i := (0.5 + |x_{i,1}|) \sigma.$$

Thus, σ controls the overall noise level, while the magnitude of the first feature determines the object-dependent noise scale. This creates heteroscedastic, object-dependent uncertainty.

Heavy-tailed noise dataset Here $x_i \sim \mathcal{N}(0, I_{10})$ and $w \sim \mathcal{N}(0, I_{10})$ are independently generated as before, but the additive noise follows a scaled Student- t distribution with three degrees of freedom:

$$y_i := w^\top x_i + \xi_i, \quad \xi_i \sim \sigma t_3.$$

Equivalently, $\xi_i := \sigma \varepsilon_i$, where $\varepsilon_i \sim t_3$. Thus, σ controls the scale of the noise, while the degrees of freedom remain fixed at three. This produces infrequent but potentially very large deviations.

Outlier contamination dataset We generate $x_i \sim \mathcal{N}(0, I_{10})$ and $w \sim \mathcal{N}(0, I_{10})$, and define

$$y_i := w^\top x_i + \xi_i,$$

where ξ_i follows a contamination model:

$$\xi_i \sim \begin{cases} \mathcal{N}(0, \sigma^2), & \text{with probability } 1 - p, \\ \mathcal{N}(0, \tau^2), & \text{with probability } p, \end{cases}$$

with $p := 0.01$ representing the outlier probability and $\tau \gg 1$ (we set $\tau := 10\sigma$) meaning that the rare errors are extremely large.

Sparse signals dataset We again draw $x_i \sim \mathcal{N}(0, I_{10})$, but the true vector of coefficients is sparse. Let s be the sparsity level (in our experiments we set $s := 2$); we randomly select a support set $S \subset \{1, \dots, 10\}$ with $|S| = s$ and define

$$w_j := \begin{cases} \mathcal{N}(0, 1), & j \in S, \\ 0, & j \notin S. \end{cases}$$

The labels follow

$$y_i := w^\top x_i + \xi_i, \quad \xi_i \sim \mathcal{N}(0, \sigma^2).$$

Covariate shift dataset This scenario uses a distribution-defined train–test split rather than randomly partitioning a common IID sample. Of the n observations, 80% are independently generated as training objects from

$$x_i^{(\text{train})} \sim \mathcal{N}(0, I_{10}),$$

while the remaining 20% are independently generated as test objects from

$$x_i^{(\text{test})} \sim \mathcal{N}(\mathbf{1}, I_{10}),$$

where $\mathbf{1}$ denotes the 10-dimensional vector of ones. A single coefficient vector $w \sim \mathcal{N}(0, I_{10})$ generates the labels in both splits:

$$y_i^{(s)} := w^\top x_i^{(s)} + \xi_i^{(s)}, \quad \xi_i^{(s)} \sim \mathcal{N}(0, \sigma^2), \quad s \in \{\text{train}, \text{test}\}.$$

The features in both splits are subsequently standardized using the means and standard deviations estimated from the training objects. Thus, the conditional distribution $p(y | x)$ is unchanged across the two splits, whereas the marginal object distribution $p(x)$ differs. Because the training and test observations are not identically distributed, the IID assumptions underlying our validity result do not hold in this scenario. We therefore include it only as a robustness stress test under distribution shift.

Friedman’s datasets We also show results for Friedman’s three synthetic datasets (Friedman, 1991) as numbered by Breiman (Breiman, 1996, Sect. 3 and Appendix B) and implemented in `scikit-learn`. We use the `scikit-learn` implementation with the default values of parameters except for the size (namely, the default number of features is 10 for Friedman 1).

For each dataset we randomly split the data into 80% training and 20% testing, standardize the features based on training statistics, and repeat all experiments across multiple (namely, 100) random seeds to reduce variance. (The covariate shift dataset is an exception in that splits are not random.) At the end of the section we also briefly discuss results for the other combinations of $n \in \{1000, 10000\}$ and $\sigma \in \{1, 3\}$.

We compare our methods against seven widely used regression baselines implemented in `scikit-learn`, including linear regressors (Linear Regression, Ridge, Lasso, Elastic Net), kernel-based methods (Support Vector Regression, SVR, with the RBF kernel), and tree-based ensembles (Random Forest and Gradient Boosting). Linear Regression in fact implements Least Squares, and Ridge (or Ridge Regression), Lasso, and Elastic Net add various elements of regularization. All models are trained with default or widely adopted hyperparameters to reflect typical practitioner usage. Performance is measured using root mean squared error (RMSE) on the held-out test set. We report mean RMSE across repeated trials for each scenario to assess their accuracy.

The overall procedure can be summarized as follows. For each dataset and random seed, we first split the data into a training portion (80%) and a held-out test portion (20%). All model fitting, calibration, and cross-fitting are performed only on the training portion. Final metrics (such as RMSE) are evaluated only on the held-out test portion. The uncalibrated baseline labelled “none” in our tables is trained once on the full training portion. The CVAR methods use the same training portion but internally perform $K = 10$ fold cross-fitting: each fold is used as a calibration set while the remaining folds are used as proper training data. The fold-specific IVAR intervals are merged into point predictions as described in Sect. 6 and averaged across folds.

Tables 1–11 show the average RMSE over 100 trials for each synthetic dataset with $\sigma = 3$ and $n = 10000$ examples and for each base algorithm without calibration (*none*) or calibrated using our CVAR method with 10 folds and parameters $m \in \{1, 10\}$ (denoted *CVAR1* and *CVAR10*, respectively, in the tables; using $m = 100$ never leads to improvements as compared with the smaller m in our experiments). Apart from the average RMSE we also report the standard error of the mean (SEM) obtained by dividing the sample standard deviation of the RMSE over the 100 trials by $\sqrt{100} = 10$; therefore, each cell in our tables has the form

$$\text{average RMSE} \pm \text{SEM}.$$

These intervals reassure us that most of our comparisons are not unduly affected by randomness, but in our summaries we only use the averages. The smallest average in each row of our tables is shown in boldface. For each table we also report the average of each column, which is the average of the average RMSE over the seven regression algorithms.

Remark 6 The results for Linear Regression and Ridge coincide in Tables 1–11, but they are sometimes slightly different in Appendix B. The reason for this closeness is that our

Table 1: Bounded Logistic dataset

	none	CVAR1	CVAR10
Elastic Net	3.809 ± 0.015	3.205 ± 0.007	3.206 ± 0.007
Gradient Boosting	3.239 ± 0.010	3.144 ± 0.006	3.145 ± 0.006
Lasso	4.001 ± 0.018	3.588 ± 0.017	3.589 ± 0.017
Linear Regression	3.257 ± 0.012	3.011 ± 0.005	3.012 ± 0.005
Random Forest	3.231 ± 0.008	3.208 ± 0.007	3.209 ± 0.007
Ridge	3.257 ± 0.012	3.011 ± 0.005	3.012 ± 0.005
SVR (RBF)	3.143 ± 0.007	3.070 ± 0.005	3.071 ± 0.005
average	3.420	3.177	3.178

Table 2: Linear Gaussian dataset

	none	CVAR1	CVAR10
Elastic Net	3.510 ± 0.015	3.147 ± 0.006	3.155 ± 0.006
Gradient Boosting	3.073 ± 0.006	3.083 ± 0.006	3.091 ± 0.006
Lasso	3.707 ± 0.018	3.419 ± 0.014	3.424 ± 0.014
Linear Regression	3.002 ± 0.005	3.019 ± 0.005	3.029 ± 0.005
Random Forest	3.138 ± 0.006	3.150 ± 0.007	3.158 ± 0.007
Ridge	3.002 ± 0.005	3.019 ± 0.005	3.029 ± 0.005
SVR (RBF)	3.087 ± 0.005	3.101 ± 0.006	3.108 ± 0.006
average	3.217	3.134	3.142

features are constructed as independent, so multicollinearity is unlikely, and the sample size is large relative to the feature count; this makes Least Squares stable, and so the default ridge penalty has relatively little impact.

The last rows in Tables 1–11 show that both of our methods, CVAR1 and CVAR10, attain better RMSE scores on average when compared with uncalibrated algorithms, especially for the bounded logistic, linear Gaussian, nonlinear, covariate shift, Friedman 1, and Friedman 2 datasets (the results for the last dataset, however, are very irregular, and in two cases our methods significantly lower the quality of predictions). In many rows the base algorithm performs better, but in a typical row of a typical table either our algorithms perform better or they lose little. For the “almost bounded” Bounded Logistic dataset our methods improve the base predictions in all rows.

We report the results for $\sigma \in \{1, 3\}$ and $n = 1000$ and for $\sigma = 1$ and $n = 10000$ settings in Appendix B. The results for $n = 10000$ and $\sigma = 1$ are similar to the results in this section, while the results for $n = 1000$ are mixed for $\sigma = 3$ and weaker for our methods in the case of low noise, $\sigma = 1$. It appears that our methods work well for larger datasets.

Table 3: Nonlinear dataset

	none	CVAR1	CVAR10
Elastic Net	3.718 ± 0.015	3.327 ± 0.006	3.336 ± 0.006
Gradient Boosting	3.100 ± 0.006	3.112 ± 0.006	3.124 ± 0.007
Lasso	3.904 ± 0.018	3.590 ± 0.014	3.596 ± 0.014
Linear Regression	3.199 ± 0.005	3.211 ± 0.006	3.221 ± 0.006
Random Forest	3.194 ± 0.008	3.210 ± 0.008	3.220 ± 0.008
Ridge	3.199 ± 0.005	3.211 ± 0.006	3.221 ± 0.006
SVR (RBF)	3.139 ± 0.006	3.157 ± 0.006	3.167 ± 0.006
average	3.350	3.260	3.269

Table 4: Heteroscedastic noise dataset

	none	CVAR1	CVAR10
Elastic Net	4.654 ± 0.016	4.386 ± 0.011	4.392 ± 0.011
Gradient Boosting	4.371 ± 0.012	4.367 ± 0.012	4.372 ± 0.012
Lasso	4.804 ± 0.017	4.587 ± 0.015	4.590 ± 0.015
Linear Regression	4.289 ± 0.011	4.295 ± 0.011	4.302 ± 0.011
Random Forest	4.430 ± 0.012	4.425 ± 0.012	4.429 ± 0.012
Ridge	4.289 ± 0.011	4.295 ± 0.011	4.302 ± 0.011
SVR (RBF)	4.358 ± 0.011	4.365 ± 0.011	4.369 ± 0.012
average	4.456	4.388	4.394

Table 5: Heavy-tailed noise dataset

	none	CVAR1	CVAR10
Elastic Net	5.444 ± 0.076	5.218 ± 0.077	5.210 ± 0.077
Gradient Boosting	5.207 ± 0.077	5.212 ± 0.077	5.204 ± 0.077
Lasso	5.578 ± 0.075	5.407 ± 0.075	5.391 ± 0.075
Linear Regression	5.120 ± 0.078	5.137 ± 0.078	5.132 ± 0.078
Random Forest	5.280 ± 0.077	5.283 ± 0.077	5.267 ± 0.077
Ridge	5.120 ± 0.078	5.137 ± 0.078	5.132 ± 0.078
SVR (RBF)	5.173 ± 0.077	5.196 ± 0.077	5.184 ± 0.077
average	5.275	5.227	5.217

Table 6: Outlier contamination dataset

	none	CVAR1	CVAR10
Elastic Net	4.674 ± 0.040	4.401 ± 0.042	4.400 ± 0.042
Gradient Boosting	4.389 ± 0.042	4.397 ± 0.042	4.388 ± 0.042
Lasso	4.826 ± 0.039	4.617 ± 0.040	4.604 ± 0.040
Linear Regression	4.290 ± 0.043	4.309 ± 0.043	4.311 ± 0.043
Random Forest	4.456 ± 0.042	4.456 ± 0.041	4.443 ± 0.041
Ridge	4.290 ± 0.043	4.309 ± 0.043	4.311 ± 0.043
SVR (RBF)	4.353 ± 0.042	4.378 ± 0.042	4.370 ± 0.042
average	4.468	4.409	4.404

Table 7: Sparse signals dataset

	none	CVAR1	CVAR10
Elastic Net	3.049 ± 0.007	2.999 ± 0.005	3.000 ± 0.005
Gradient Boosting	3.009 ± 0.005	3.008 ± 0.005	3.008 ± 0.005
Lasso	3.079 ± 0.009	3.026 ± 0.006	3.027 ± 0.006
Linear Regression	2.995 ± 0.005	2.998 ± 0.005	2.998 ± 0.005
Random Forest	3.046 ± 0.005	3.027 ± 0.005	3.026 ± 0.005
Ridge	2.995 ± 0.005	2.998 ± 0.005	2.998 ± 0.005
SVR (RBF)	3.038 ± 0.005	3.025 ± 0.005	3.024 ± 0.005
average	3.030	3.012	3.012

Table 8: Covariate shift dataset

	none	CVAR1	CVAR10
Elastic Net	3.887 ± 0.050	3.352 ± 0.023	3.378 ± 0.025
Gradient Boosting	3.213 ± 0.011	3.275 ± 0.019	3.306 ± 0.023
Lasso	4.191 ± 0.065	3.809 ± 0.044	3.825 ± 0.045
Linear Regression	3.007 ± 0.004	3.109 ± 0.018	3.143 ± 0.021
Random Forest	3.371 ± 0.019	3.426 ± 0.024	3.453 ± 0.027
Ridge	3.007 ± 0.004	3.109 ± 0.018	3.143 ± 0.021
SVR (RBF)	3.600 ± 0.043	3.658 ± 0.046	3.675 ± 0.048
average	3.468	3.391	3.417

Table 9: Friedman 1 dataset

	none	CVAR1	CVAR10
Elastic Net	4.383 ± 0.008	3.871 ± 0.007	3.879 ± 0.007
Gradient Boosting	3.139 ± 0.006	3.173 ± 0.006	3.190 ± 0.006
Lasso	4.353 ± 0.008	3.960 ± 0.007	3.967 ± 0.007
Linear Regression	3.864 ± 0.007	3.858 ± 0.007	3.865 ± 0.007
Random Forest	3.276 ± 0.006	3.312 ± 0.006	3.328 ± 0.006
Ridge	3.864 ± 0.007	3.858 ± 0.007	3.865 ± 0.007
SVR (RBF)	3.226 ± 0.006	3.258 ± 0.006	3.273 ± 0.006
average	3.729	3.613	3.624

Table 10: Friedman 2 dataset

	none	CVAR1	CVAR10
Elastic Net	181.783 ± 0.340	82.684 ± 0.170	84.122 ± 0.168
Gradient Boosting	15.432 ± 0.057	48.841 ± 0.119	51.088 ± 0.149
Lasso	137.838 ± 0.233	82.678 ± 0.171	84.126 ± 0.169
Linear Regression	137.820 ± 0.233	82.720 ± 0.170	84.167 ± 0.169
Random Forest	6.550 ± 0.019	47.315 ± 0.119	49.651 ± 0.151
Ridge	137.820 ± 0.233	82.720 ± 0.170	84.167 ± 0.169
SVR (RBF)	140.878 ± 0.536	105.137 ± 0.376	105.298 ± 0.397
average	108.303	76.013	77.517

Table 11: Friedman 3 dataset

	none	CVAR1	CVAR10
Elastic Net	3.022 ± 0.005	3.022 ± 0.005	3.022 ± 0.005
Gradient Boosting	3.023 ± 0.005	3.020 ± 0.005	3.018 ± 0.005
Lasso	3.022 ± 0.005	3.022 ± 0.005	3.022 ± 0.005
Linear Regression	3.013 ± 0.005	3.016 ± 0.005	3.014 ± 0.005
Random Forest	3.111 ± 0.005	3.026 ± 0.005	3.024 ± 0.005
Ridge	3.013 ± 0.005	3.016 ± 0.005	3.014 ± 0.005
SVR (RBF)	3.023 ± 0.005	3.020 ± 0.005	3.018 ± 0.005
average	3.032	3.020	3.019

Table 12: Regression interval widths for the naive IVAR, IVAR1, and IVAR10 as described in text

Dataset	naive IVAR	IVAR1	IVAR10
Bounded Logistic	0.158	0.146	0.124
Linear Gaussian	0.215	0.195	0.157
Heavy-tailed noise	0.309	0.208	0.130
Outlier contamination	0.423	0.263	0.120
Friedman 1	0.227	0.206	0.173

In Appendix B we also give results for four real-life datasets. Our methods tend to improve the performance of the base algorithms unless the dataset is small, but the tendency is weak.

The widths of regression intervals

In the main part of this section we only studied the performance of our point predictors, CVAR1 and CVAR10, ultimately based on inductive Venn–Abers prediction but not having any formal properties of validity. The reason for this choice was that the properties of validity that we established for IVARs are very different from those used in existing literature, and any comparisons with existing algorithms would be of a very limited interest. For example, the main algorithm (Algorithm 2) in [van der Laan and Alaa \(2024\)](#) outputs prediction rather than regression intervals and, therefore, produces intervals that are wider than ours by orders of magnitude for a typical dataset considered earlier in this section. It does not mean at all that our algorithms are better since, unlike Algorithm 2 in [van der Laan and Alaa \(2024\)](#), they do not guarantee a good coverage probability.

Table 12 compares the widths of regression intervals produced by IVAR1 and IVAR10, i.e., the IVAR with $m = 1$ and $m = 10$, respectively. Even though such comparison is internal to this paper, it is somewhat instructive as it shows that IVAR10 produces regression intervals that are considerably shorter. The column “naive IVAR” in the table gives results for the modification of the IVAR in which y_* and y^* are replaced by the smallest and largest labels, respectively, in the calibration set (as suggested by a reviewer); these results are predictably even looser than those for IVAR1. Table 12 gives results only for a subset of datasets considered in this section, but the results are typical; they are for the Gradient Boosting base regressor, $N := 10000$, and $\sigma := 3$. The widths of the regression intervals are computed from the training set only; the training set is split randomly into $K = 10$ folds and each fold in turn is used as the calibration set (similarly to CVARs); the widths of the resulting regression intervals are then averaged.

9. Conclusion

This paper presents new validity guarantees for regression estimation. Computational experiments show that this leads to a limited improvement in the performance of standard point regressors on large datasets.

These are some directions of further research.

- An alternative merging procedure to the ones used in Sect. 7 is to try and merge the regression intervals coming from the K folds directly (in a minimax manner, as in Vovk et al. 2022, Sect. 6.4.5) in order to obtain an overall regression estimate, without the intermediate step of merging each regression interval. It would be interesting to compare it experimentally with the procedure used in this paper.
- In this paper we explore the predictive efficiency of the CVAR in simulation and empirical studies. Alternatively, we could try and prove theoretical results along the lines of the Burnaev–Wasserman programme, as described in Vovk et al. (2022, Sects. 2.5 and 2.9.7). As a first step, we may assume that the base predictions r coincide with the true regression function, $r := \mathbb{E}(Y \mid X = x)$.
- Our methods tend to improve significantly the quality of predictions made by Lasso and Elastic Net. A very common pattern of the boldface (i.e., best) entries in our tables is “CVAR1, none, CVAR1, none, none, none, none” (from top to bottom); we can observe it in three tables (Tables 2, 3, and 8) in the main paper, and in more than half of the tables (20 out of 37) in Appendix B. For this pattern our methods help only for Lasso and Elastic Net. However, it can be argued that applying our methods destroys, at least partially, a valued property of Lasso and Elastic Net, automatic feature selection. Optimizing the number of features is another interesting desideratum, alongside predictive efficiency.
- On the other hand, our methods do not consistently improve on bagging (e.g., CVAR10 improves on Random Forest 6 times out of 11 in Tables 1–11, and for CVAR1 the results are even worse). One of the reviewers suggested that there is an element of bagging in cross Venn–Abers regression as we average point predictions coming from different folds. Such averaging does not help for a regressor that has already been bagged. It would be interesting to disentangle the effects of bagging and calibration in CVARs’ performance.
- In our empirical studies we focused on point predictors, CVARs, derived from IVARs but not enjoying any formal properties of validity. Only in Table 12 did we compare the widths of regression intervals coming from IVAR1 and IVAR10. Perhaps the most important direction of further research is the empirical study of IVARs, both their validity and efficiency. While validity is asserted in Theorem 4, checking it empirically might shed further light on its meaning. Exploring the efficiency of IVARs requires designing suitable ways of measuring it, along the lines of proper scoring rules (see, e.g., Dawid 2006) in probability forecasting and conditionally proper criteria of efficiency (Vovk et al., 2022, Sect. 3.1.5) in conformal prediction. A natural question is how to balance encouraging short regression intervals and a good performance of, say, their midpoints (e.g., in the sense of RMSE).

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Appendix A. Proofs

A.1. Proof of Theorem 2

The selector S that we use for demonstrating the theorem is a modification of steps 4–7 in the description of the bounded IVAR corresponding to using the true test label. Namely, we fit isotonic regression to $(r_1, y_1), \dots, (r_k, y_k), (r, y)$, where y is the true label of the test object, obtaining an isotonic calibrator f . Set $S := f(r)$.

First we need to check that S is indeed a selector, namely that $f_*(r) \leq S \leq f^*(r)$. This follows from $C_* \leq y \leq C^*$ and the monotonicity property of PAVA given by the following lemma.

Lemma 7 *If $y_i \leq y'_i$ for all $i \in \{1, \dots, n\}$, then $f \leq f'$, where f (resp. f') is the isotonic regression fitted to $(r_1, y_1), \dots, (r_n, y_n)$ (resp. to $(r_1, y'_1), \dots, (r_n, y'_n)$).*

Proof This is an immediate corollary of the max-min formulas (as given in, e.g., [Barlow et al. 1972](#), p. 19). ■

Now let us check that $\mathbb{E}(Y \mid S) = S$ even conditionally on the bag $B = \{Z_1, \dots, Z_k, Z\}$ of k calibration examples and a test example under the uniform probability measure on all $(k+1)!$ orderings of the bag (see [Vovk et al. 2022](#), Lemma A.3). The value of S determines the solution block containing the test example. Therefore, both S and $\mathbb{E}(Y \mid S, B)$ will equal the arithmetic mean of the labels in that solution block.

A.2. Proof of Theorem 4

Fix an IVAR. The selector S is produced by the following ideal picture. Let y be the true label of the test object x . Then the selector S associated with the IVAR is defined by the following recipe applied after splitting the training set and training the base regression algorithm on the proper training set (which gives us a prediction rule).

1. Winsorize the labels, namely:

- replace the m largest labels in the *augmented calibration sequence*

$$(x_1, y_1), \dots, (x_k, y_k), (x, y)$$

by the $(m+1)$ th largest label;

- replace the m smallest labels in the augmented calibration sequence by the $(m+1)$ th smallest label.

2. Find the base predictions $r_1, \dots, r_k, r$ of the objects $x_1, \dots, x_k, x$ using the prediction rule.
3. Fit isotonic regression f to $(r_1, y_1), \dots, (r_k, y_k), (r, y)$ (with the Winsorized labels).
4. Set $S := f(r)$.

Step 1 regularizes the labels moderating the $2m$ most extreme ones. The recipe treats all elements of the augmented calibration sequence symmetrically, which is essential in the proof of $\mathbb{E}(Y' \mid S) = S$; let us call it the *symmetric recipe*.

First let us check that S is indeed a selector for the IVAR. Consider three cases:

- If y is one of the m largest labels in the augmented calibration sequence, the definition of IVAR and the symmetric recipe modify the labels in the same way; in particular, the $m-1$ largest calibration labels and the test label are replaced by y^* , and the m smallest calibration labels are replaced by y_* . In this case $S = \hat{y}^*$.
- Analogously, if y is one of the m smallest labels in the augmented calibration sequence, we have $S = \hat{y}_*$.
- Otherwise, the monotonicity property of isotonic regression given in Lemma 7 above implies that we still have $S \in [\hat{y}_*, \hat{y}^*]$.

In all three cases, $S \in [\hat{y}_*, \hat{y}^*]$, which means that S is a selector.

Table 13: Bounded Logistic dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.551 ± 0.021	1.506 ± 0.012	1.508 ± 0.012
Gradient Boosting	1.532 ± 0.018	1.323 ± 0.009	1.326 ± 0.009
Lasso	2.830 ± 0.024	2.201 ± 0.025	2.203 ± 0.025
Linear Regression	1.596 ± 0.021	1.031 ± 0.002	1.035 ± 0.002
Random Forest	1.497 ± 0.017	1.463 ± 0.014	1.466 ± 0.014
Ridge	1.596 ± 0.021	1.031 ± 0.002	1.035 ± 0.002
SVR (RBF)	1.211 ± 0.009	1.093 ± 0.002	1.097 ± 0.002
average	1.830	1.378	1.382

Remark 8 The first two cases considered in the proof demonstrate the tightness, in some sense, of the regression interval $[\hat{y}_*, \hat{y}^*]$.

It remains to prove $\mathbb{E}(Y' \mid S) = S$. We prove this equality conditionally on the set of all permutations (equiprobable) of a given augmented calibration sequence, where a permutation π of $\{1, \dots, k+1\}$ makes $z_{\pi^{-1}(k+1)}$ the test example. The Winsorized test label (3) over the permutations is the same random variable as the test label produced according to the symmetric recipe applied to the permutations. It remains to remember that the isotonic regression at the base prediction r for the test object is the mean of the labels in the augmented calibration sequence in the same solution block as the test example (Barlow et al., 1972, pp. 13–15). Solution blocks are level sets of the isotonic regression, which implies S being the conditional average of the Winsorized test label given S .

Appendix B. Further Experimental Results

In Tables 13–23 we show the experimental results in the case of $\sigma = 1$ parallel to those reported in Sect. 8 of the main paper; in particular, we still have $n = 10000$. Our comments in the main paper are also applicable here. In particular, it is still true that on average both CVAR1 and CVAR10 produce better results than the base algorithms in all 11 tables. The level of noise does not appear to be a crucial parameter.

Tables 24–34 give results for $n = 1000$ examples and noise level $\sigma = 3$. For these smaller datasets the results are mixed, as we said in the main paper.

In Tables 35–45 we have $n = 1000$ examples, but the noise level is smaller, $\sigma = 1$. The results for our methods appear even worse than for the more challenging case $\sigma = 3$; the base algorithms work best on average in nine cases out of eleven.

Finally, in Tables 46–49 we show results for the following four real-life datasets: Bias correction (in full, Bias correction of numerical prediction model temperature forecast) (Cho et al., 2020) with 7750 examples and 21 features, Wine quality (Cortez et al., 2009) with 6497 examples and 12 features (including colour, red or white), Airfoil self-noise (Brooks et al., 1989) with 1503 examples and 5 features, and Student performance (Word et al., 1990) with 649 examples and 30 features. The results are mixed; in two cases, our methods

Table 14: Linear Gaussian dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.079 ± 0.024	1.395 ± 0.009	1.414 ± 0.010
Gradient Boosting	1.155 ± 0.008	1.189 ± 0.008	1.213 ± 0.009
Lasso	2.397 ± 0.026	1.921 ± 0.024	1.933 ± 0.024
Linear Regression	1.001 ± 0.002	1.084 ± 0.005	1.110 ± 0.006
Random Forest	1.281 ± 0.013	1.312 ± 0.014	1.333 ± 0.014
Ridge	1.001 ± 0.002	1.084 ± 0.005	1.110 ± 0.006
SVR (RBF)	1.094 ± 0.004	1.155 ± 0.006	1.179 ± 0.007
average	1.430	1.306	1.327

 Table 15: Nonlinear dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.417 ± 0.021	1.763 ± 0.007	1.781 ± 0.008
Gradient Boosting	1.227 ± 0.009	1.260 ± 0.009	1.289 ± 0.010
Lasso	2.695 ± 0.024	2.212 ± 0.020	2.223 ± 0.020
Linear Regression	1.494 ± 0.003	1.539 ± 0.005	1.560 ± 0.005
Random Forest	1.417 ± 0.015	1.455 ± 0.015	1.479 ± 0.016
Ridge	1.494 ± 0.003	1.539 ± 0.005	1.560 ± 0.005
SVR (RBF)	1.202 ± 0.004	1.272 ± 0.006	1.301 ± 0.007
average	1.707	1.577	1.599

 Table 16: Heteroscedastic noise dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.315 ± 0.022	1.722 ± 0.008	1.740 ± 0.009
Gradient Boosting	1.550 ± 0.007	1.570 ± 0.007	1.591 ± 0.008
Lasso	2.604 ± 0.025	2.173 ± 0.021	2.184 ± 0.021
Linear Regression	1.430 ± 0.004	1.480 ± 0.005	1.504 ± 0.006
Random Forest	1.640 ± 0.010	1.662 ± 0.011	1.681 ± 0.011
Ridge	1.430 ± 0.004	1.480 ± 0.005	1.504 ± 0.006
SVR (RBF)	1.509 ± 0.005	1.546 ± 0.006	1.566 ± 0.007
average	1.782	1.662	1.681

Table 17: Heavy-tailed noise dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.512 ± 0.029	1.957 ± 0.025	1.981 ± 0.025
Gradient Boosting	1.819 ± 0.026	1.830 ± 0.026	1.856 ± 0.026
Lasso	2.785 ± 0.030	2.381 ± 0.029	2.395 ± 0.029
Linear Regression	1.707 ± 0.026	1.742 ± 0.026	1.772 ± 0.026
Random Forest	1.901 ± 0.026	1.915 ± 0.026	1.938 ± 0.026
Ridge	1.707 ± 0.026	1.742 ± 0.026	1.772 ± 0.026
SVR (RBF)	1.770 ± 0.025	1.797 ± 0.025	1.822 ± 0.025
average	2.028	1.909	1.934

Table 18: Outlier contamination dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.330 ± 0.022	1.718 ± 0.014	1.748 ± 0.014
Gradient Boosting	1.561 ± 0.013	1.573 ± 0.014	1.607 ± 0.014
Lasso	2.620 ± 0.025	2.182 ± 0.022	2.199 ± 0.022
Linear Regression	1.430 ± 0.014	1.473 ± 0.014	1.512 ± 0.014
Random Forest	1.656 ± 0.015	1.671 ± 0.015	1.700 ± 0.015
Ridge	1.430 ± 0.014	1.473 ± 0.014	1.512 ± 0.014
SVR (RBF)	1.500 ± 0.014	1.533 ± 0.014	1.567 ± 0.014
average	1.790	1.661	1.692

Table 19: Sparse signals dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	1.144 ± 0.012	1.016 ± 0.003	1.019 ± 0.003
Gradient Boosting	1.004 ± 0.002	1.010 ± 0.002	1.012 ± 0.002
Lasso	1.214 ± 0.018	1.085 ± 0.011	1.087 ± 0.011
Linear Regression	0.998 ± 0.002	1.005 ± 0.002	1.008 ± 0.002
Random Forest	1.020 ± 0.002	1.022 ± 0.002	1.025 ± 0.003
Ridge	0.998 ± 0.002	1.005 ± 0.002	1.008 ± 0.002
SVR (RBF)	1.030 ± 0.002	1.029 ± 0.003	1.032 ± 0.003
average	1.058	1.025	1.027

Table 20: Covariate shift dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.619 ± 0.068	1.794 ± 0.040	1.841 ± 0.043
Gradient Boosting	1.462 ± 0.024	1.590 ± 0.037	1.646 ± 0.042
Lasso	3.047 ± 0.083	2.537 ± 0.063	2.563 ± 0.064
Linear Regression	1.002 ± 0.001	1.312 ± 0.036	1.377 ± 0.041
Random Forest	1.778 ± 0.036	1.863 ± 0.042	1.908 ± 0.046
Ridge	1.002 ± 0.001	1.312 ± 0.036	1.377 ± 0.041
SVR (RBF)	1.888 ± 0.052	2.025 ± 0.060	2.058 ± 0.062
average	1.828	1.776	1.824

 Table 21: Friedman 1 dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	3.344 ± 0.005	2.649 ± 0.005	2.658 ± 0.005
Gradient Boosting	1.307 ± 0.004	1.418 ± 0.004	1.447 ± 0.004
Lasso	3.305 ± 0.005	2.775 ± 0.005	2.783 ± 0.005
Linear Regression	2.629 ± 0.005	2.628 ± 0.005	2.637 ± 0.005
Random Forest	1.498 ± 0.003	1.603 ± 0.003	1.628 ± 0.003
Ridge	2.629 ± 0.005	2.628 ± 0.005	2.637 ± 0.005
SVR (RBF)	1.309 ± 0.003	1.428 ± 0.003	1.458 ± 0.003
average	2.289	2.161	2.178

 Table 22: Friedman 2 dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	181.762 ± 0.340	82.637 ± 0.170	84.073 ± 0.168
Gradient Boosting	15.208 ± 0.061	48.795 ± 0.120	51.037 ± 0.150
Lasso	137.802 ± 0.233	82.631 ± 0.171	84.076 ± 0.170
Linear Regression	137.783 ± 0.233	82.675 ± 0.171	84.118 ± 0.170
Random Forest	5.791 ± 0.021	47.239 ± 0.119	49.570 ± 0.152
Ridge	137.784 ± 0.233	82.675 ± 0.171	84.118 ± 0.170
SVR (RBF)	140.800 ± 0.538	105.034 ± 0.376	105.190 ± 0.398
average	108.133	75.955	77.455

Table 23: Friedman 3 dataset ($\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	1.050 ± 0.002	1.050 ± 0.002	1.050 ± 0.002
Gradient Boosting	1.009 ± 0.002	1.009 ± 0.002	1.009 ± 0.002
Lasso	1.050 ± 0.002	1.050 ± 0.002	1.050 ± 0.002
Linear Regression	1.022 ± 0.002	1.014 ± 0.002	1.014 ± 0.002
Random Forest	1.042 ± 0.002	1.021 ± 0.002	1.020 ± 0.002
Ridge	1.022 ± 0.002	1.014 ± 0.002	1.014 ± 0.002
SVR (RBF)	1.012 ± 0.002	1.011 ± 0.002	1.010 ± 0.002
average	1.030	1.024	1.024

Table 24: Bounded Logistic dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	3.831 ± 0.021	3.302 ± 0.018	3.417 ± 0.018
Gradient Boosting	3.420 ± 0.019	3.408 ± 0.016	3.514 ± 0.016
Lasso	4.035 ± 0.023	3.691 ± 0.023	3.773 ± 0.022
Linear Regression	3.288 ± 0.019	3.120 ± 0.015	3.247 ± 0.015
Random Forest	3.449 ± 0.018	3.466 ± 0.017	3.564 ± 0.017
Ridge	3.288 ± 0.019	3.120 ± 0.015	3.247 ± 0.015
SVR (RBF)	3.345 ± 0.018	3.242 ± 0.016	3.358 ± 0.016
average	3.522	3.336	3.446

Table 25: Linear Gaussian dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	3.549 ± 0.022	3.289 ± 0.019	3.431 ± 0.023
Gradient Boosting	3.228 ± 0.016	3.342 ± 0.019	3.475 ± 0.024
Lasso	3.755 ± 0.025	3.562 ± 0.023	3.661 ± 0.026
Linear Regression	3.016 ± 0.015	3.173 ± 0.018	3.333 ± 0.022
Random Forest	3.289 ± 0.018	3.375 ± 0.022	3.502 ± 0.026
Ridge	3.016 ± 0.015	3.173 ± 0.018	3.333 ± 0.022
SVR (RBF)	3.256 ± 0.019	3.307 ± 0.020	3.438 ± 0.025
average	3.301	3.317	3.453

Table 26: Nonlinear dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	3.749 ± 0.023	3.473 ± 0.019	3.626 ± 0.024
Gradient Boosting	3.265 ± 0.017	3.418 ± 0.020	3.579 ± 0.025
Lasso	3.941 ± 0.026	3.730 ± 0.023	3.841 ± 0.027
Linear Regression	3.210 ± 0.014	3.366 ± 0.018	3.536 ± 0.023
Random Forest	3.365 ± 0.019	3.483 ± 0.022	3.634 ± 0.028
Ridge	3.209 ± 0.014	3.366 ± 0.018	3.536 ± 0.023
SVR (RBF)	3.365 ± 0.019	3.425 ± 0.020	3.580 ± 0.025
average	3.444	3.466	3.619

 Table 27: Heteroscedastic noise dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	4.677 ± 0.032	4.474 ± 0.031	4.599 ± 0.033
Gradient Boosting	4.545 ± 0.031	4.580 ± 0.031	4.682 ± 0.033
Lasso	4.837 ± 0.033	4.688 ± 0.033	4.771 ± 0.035
Linear Regression	4.282 ± 0.029	4.382 ± 0.030	4.524 ± 0.032
Random Forest	4.522 ± 0.031	4.572 ± 0.032	4.675 ± 0.034
Ridge	4.282 ± 0.029	4.382 ± 0.030	4.524 ± 0.032
SVR (RBF)	4.479 ± 0.031	4.498 ± 0.031	4.608 ± 0.033
average	4.518	4.511	4.626

 Table 28: Heavy-tailed noise dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	5.266 ± 0.091	5.094 ± 0.092	5.188 ± 0.092
Gradient Boosting	5.297 ± 0.095	5.208 ± 0.093	5.280 ± 0.093
Lasso	5.409 ± 0.089	5.279 ± 0.091	5.336 ± 0.091
Linear Regression	4.949 ± 0.093	5.018 ± 0.093	5.129 ± 0.093
Random Forest	5.237 ± 0.093	5.207 ± 0.094	5.277 ± 0.093
Ridge	4.949 ± 0.093	5.018 ± 0.093	5.129 ± 0.093
SVR (RBF)	5.084 ± 0.093	5.107 ± 0.093	5.191 ± 0.093
average	5.170	5.133	5.218

Table 29: Outlier contamination dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	4.561 ± 0.102	4.358 ± 0.105	4.469 ± 0.103
Gradient Boosting	4.488 ± 0.108	4.457 ± 0.106	4.542 ± 0.104
Lasso	4.727 ± 0.099	4.591 ± 0.103	4.652 ± 0.101
Linear Regression	4.171 ± 0.108	4.268 ± 0.107	4.401 ± 0.105
Random Forest	4.480 ± 0.107	4.479 ± 0.105	4.555 ± 0.103
Ridge	4.171 ± 0.108	4.269 ± 0.107	4.401 ± 0.105
SVR (RBF)	4.329 ± 0.106	4.374 ± 0.106	4.474 ± 0.104
average	4.418	4.399	4.499

Table 30: Sparse signals dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	3.022 ± 0.018	2.982 ± 0.016	2.997 ± 0.017
Gradient Boosting	3.106 ± 0.017	3.038 ± 0.017	3.038 ± 0.018
Lasso	3.042 ± 0.018	3.009 ± 0.016	3.024 ± 0.017
Linear Regression	2.978 ± 0.016	2.995 ± 0.016	3.004 ± 0.017
Random Forest	3.064 ± 0.017	3.030 ± 0.017	3.032 ± 0.019
Ridge	2.978 ± 0.016	2.995 ± 0.016	3.004 ± 0.017
SVR (RBF)	3.041 ± 0.017	3.030 ± 0.017	3.033 ± 0.019
average	3.033	3.011	3.019

Table 31: Covariate shift dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	3.954 ± 0.064	3.693 ± 0.067	3.934 ± 0.082
Gradient Boosting	3.490 ± 0.029	3.832 ± 0.069	4.051 ± 0.084
Lasso	4.273 ± 0.080	4.054 ± 0.072	4.234 ± 0.085
Linear Regression	3.040 ± 0.018	3.536 ± 0.067	3.808 ± 0.083
Random Forest	3.624 ± 0.045	3.881 ± 0.071	4.097 ± 0.085
Ridge	3.040 ± 0.018	3.536 ± 0.067	3.808 ± 0.083
SVR (RBF)	4.151 ± 0.086	4.179 ± 0.089	4.342 ± 0.099
average	3.653	3.816	4.039

Table 32: Friedman 1 dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	4.385 ± 0.023	4.055 ± 0.021	4.269 ± 0.022
Gradient Boosting	3.378 ± 0.018	3.660 ± 0.020	3.934 ± 0.022
Lasso	4.365 ± 0.023	4.157 ± 0.021	4.359 ± 0.023
Linear Regression	3.893 ± 0.021	4.043 ± 0.021	4.256 ± 0.022
Random Forest	3.568 ± 0.019	3.808 ± 0.020	4.061 ± 0.022
Ridge	3.893 ± 0.021	4.043 ± 0.021	4.256 ± 0.022
SVR (RBF)	3.746 ± 0.021	3.865 ± 0.021	4.104 ± 0.023
average	3.890	3.947	4.177

 Table 33: Friedman 2 dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	181.546 ± 1.247	151.625 ± 1.044	181.004 ± 1.604
Gradient Boosting	23.936 ± 0.181	142.035 ± 0.940	171.828 ± 1.657
Lasso	137.739 ± 0.784	151.686 ± 1.032	181.067 ± 1.600
Linear Regression	137.743 ± 0.784	151.713 ± 1.033	181.083 ± 1.600
Random Forest	20.166 ± 0.202	141.653 ± 0.914	171.544 ± 1.636
Ridge	137.746 ± 0.784	151.711 ± 1.033	181.082 ± 1.600
SVR (RBF)	345.749 ± 2.419	170.491 ± 1.685	193.626 ± 2.036
average	140.661	151.559	180.176

 Table 34: Friedman 3 dataset ($n = 1000$ and $\sigma = 3$)

	none	CVAR1	CVAR10
Elastic Net	3.019 ± 0.015	3.019 ± 0.015	3.020 ± 0.015
Gradient Boosting	3.130 ± 0.015	3.039 ± 0.015	3.024 ± 0.015
Lasso	3.019 ± 0.015	3.019 ± 0.015	3.020 ± 0.015
Linear Regression	3.018 ± 0.015	3.036 ± 0.015	3.019 ± 0.015
Random Forest	3.146 ± 0.016	3.045 ± 0.015	3.028 ± 0.015
Ridge	3.018 ± 0.015	3.036 ± 0.015	3.019 ± 0.015
SVR (RBF)	3.054 ± 0.016	3.036 ± 0.015	3.022 ± 0.015
average	3.058	3.033	3.022

Table 35: Bounded Logistic dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.580 ± 0.022	1.749 ± 0.016	1.849 ± 0.015
Gradient Boosting	1.708 ± 0.021	1.770 ± 0.016	1.869 ± 0.015
Lasso	2.877 ± 0.026	2.390 ± 0.026	2.454 ± 0.025
Linear Regression	1.630 ± 0.023	1.380 ± 0.007	1.506 ± 0.007
Random Forest	1.870 ± 0.023	1.953 ± 0.019	2.042 ± 0.018
Ridge	1.629 ± 0.023	1.380 ± 0.007	1.506 ± 0.007
SVR (RBF)	1.578 ± 0.017	1.504 ± 0.008	1.619 ± 0.008
average	1.982	1.732	1.835

Table 36: Linear Gaussian dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.141 ± 0.029	1.744 ± 0.024	1.956 ± 0.031
Gradient Boosting	1.326 ± 0.015	1.676 ± 0.028	1.899 ± 0.035
Lasso	2.471 ± 0.032	2.189 ± 0.030	2.335 ± 0.035
Linear Regression	1.005 ± 0.005	1.529 ± 0.025	1.772 ± 0.032
Random Forest	1.566 ± 0.024	1.797 ± 0.033	2.001 ± 0.039
Ridge	1.005 ± 0.005	1.529 ± 0.025	1.772 ± 0.032
SVR (RBF)	1.418 ± 0.020	1.651 ± 0.027	1.879 ± 0.035
average	1.562	1.731	1.945

Table 37: Nonlinear dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.469 ± 0.028	2.069 ± 0.023	2.284 ± 0.031
Gradient Boosting	1.430 ± 0.016	1.835 ± 0.030	2.087 ± 0.036
Lasso	2.755 ± 0.032	2.458 ± 0.030	2.613 ± 0.035
Linear Regression	1.496 ± 0.008	1.896 ± 0.023	2.139 ± 0.031
Random Forest	1.735 ± 0.025	2.004 ± 0.033	2.230 ± 0.039
Ridge	1.496 ± 0.008	1.896 ± 0.023	2.139 ± 0.031
SVR (RBF)	1.622 ± 0.020	1.870 ± 0.028	2.118 ± 0.035
average	1.858	2.004	2.230

Table 38: Heteroscedastic noise dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.379 ± 0.027	2.014 ± 0.023	2.221 ± 0.030
Gradient Boosting	1.697 ± 0.014	1.977 ± 0.026	2.192 ± 0.033
Lasso	2.683 ± 0.030	2.422 ± 0.029	2.564 ± 0.033
Linear Regression	1.427 ± 0.010	1.826 ± 0.023	2.062 ± 0.030
Random Forest	1.875 ± 0.022	2.069 ± 0.030	2.267 ± 0.036
Ridge	1.427 ± 0.010	1.826 ± 0.023	2.062 ± 0.030
SVR (RBF)	1.777 ± 0.019	1.945 ± 0.026	2.163 ± 0.033
average	1.895	2.011	2.219

 Table 39: Heavy-tailed noise dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.492 ± 0.034	2.140 ± 0.033	2.335 ± 0.037
Gradient Boosting	1.907 ± 0.032	2.122 ± 0.036	2.321 ± 0.040
Lasso	2.783 ± 0.037	2.523 ± 0.036	2.657 ± 0.039
Linear Regression	1.650 ± 0.031	1.970 ± 0.035	2.191 ± 0.038
Random Forest	2.043 ± 0.034	2.196 ± 0.039	2.384 ± 0.042
Ridge	1.650 ± 0.031	1.970 ± 0.035	2.191 ± 0.038
SVR (RBF)	1.932 ± 0.034	2.078 ± 0.036	2.279 ± 0.039
average	2.065	2.143	2.337

 Table 40: Outlier contamination dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.329 ± 0.032	1.952 ± 0.032	2.159 ± 0.034
Gradient Boosting	1.684 ± 0.034	1.929 ± 0.036	2.138 ± 0.037
Lasso	2.634 ± 0.034	2.375 ± 0.035	2.513 ± 0.036
Linear Regression	1.390 ± 0.036	1.768 ± 0.035	2.004 ± 0.036
Random Forest	1.852 ± 0.035	2.020 ± 0.037	2.217 ± 0.039
Ridge	1.390 ± 0.036	1.768 ± 0.035	2.004 ± 0.036
SVR (RBF)	1.707 ± 0.034	1.882 ± 0.036	2.099 ± 0.037
average	1.855	1.956	2.162

Table 41: Sparse signals dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	1.153 ± 0.016	1.062 ± 0.011	1.105 ± 0.016
Gradient Boosting	1.039 ± 0.006	1.072 ± 0.011	1.112 ± 0.017
Lasso	1.204 ± 0.018	1.134 ± 0.013	1.167 ± 0.017
Linear Regression	0.993 ± 0.005	1.049 ± 0.011	1.093 ± 0.016
Random Forest	1.030 ± 0.006	1.070 ± 0.012	1.110 ± 0.017
Ridge	0.993 ± 0.005	1.049 ± 0.011	1.093 ± 0.016
SVR (RBF)	1.073 ± 0.009	1.091 ± 0.014	1.128 ± 0.019
average	1.069	1.075	1.115

Table 42: Covariate shift dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	2.706 ± 0.080	2.373 ± 0.088	2.688 ± 0.105
Gradient Boosting	1.717 ± 0.037	2.395 ± 0.090	2.705 ± 0.106
Lasso	3.149 ± 0.095	2.918 ± 0.089	3.136 ± 0.104
Linear Regression	1.013 ± 0.006	2.128 ± 0.091	2.491 ± 0.108
Random Forest	2.152 ± 0.060	2.582 ± 0.092	2.859 ± 0.107
Ridge	1.013 ± 0.006	2.128 ± 0.091	2.491 ± 0.108
SVR (RBF)	2.740 ± 0.101	2.936 ± 0.110	3.126 ± 0.120
average	2.070	2.494	2.785

Table 43: Friedman 1 dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	3.330 ± 0.015	2.926 ± 0.014	3.175 ± 0.015
Gradient Boosting	1.551 ± 0.009	2.243 ± 0.012	2.583 ± 0.015
Lasso	3.301 ± 0.016	3.051 ± 0.016	3.289 ± 0.016
Linear Regression	2.634 ± 0.014	2.907 ± 0.014	3.155 ± 0.015
Random Forest	2.039 ± 0.011	2.525 ± 0.012	2.830 ± 0.014
Ridge	2.633 ± 0.014	2.907 ± 0.014	3.155 ± 0.015
SVR (RBF)	2.172 ± 0.013	2.530 ± 0.014	2.831 ± 0.015
average	2.523	2.727	3.003

Table 44: Friedman 2 dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	181.524 ± 1.246	151.614 ± 1.044	180.970 ± 1.603
Gradient Boosting	23.714 ± 0.181	142.014 ± 0.949	171.767 ± 1.660
Lasso	137.715 ± 0.782	151.681 ± 1.031	181.033 ± 1.599
Linear Regression	137.720 ± 0.782	151.707 ± 1.032	181.048 ± 1.599
Random Forest	19.889 ± 0.206	141.642 ± 0.918	171.495 ± 1.637
Ridge	137.723 ± 0.782	151.706 ± 1.032	181.049 ± 1.599
SVR (RBF)	345.751 ± 2.424	170.534 ± 1.669	193.720 ± 2.034
average	140.576	151.557	180.155

 Table 45: Friedman 3 dataset ($n = 1000$ and $\sigma = 1$)

	none	CVAR1	CVAR10
Elastic Net	1.048 ± 0.005	1.048 ± 0.005	1.048 ± 0.005
Gradient Boosting	1.045 ± 0.005	1.029 ± 0.005	1.028 ± 0.005
Lasso	1.048 ± 0.005	1.048 ± 0.005	1.048 ± 0.005
Linear Regression	1.024 ± 0.005	1.021 ± 0.005	1.022 ± 0.005
Random Forest	1.054 ± 0.005	1.031 ± 0.005	1.030 ± 0.005
Ridge	1.024 ± 0.005	1.021 ± 0.005	1.022 ± 0.005
SVR (RBF)	1.035 ± 0.005	1.026 ± 0.005	1.026 ± 0.005
average	1.040	1.032	1.032

slightly improve the performance of the base algorithms on average. (One of the datasets where our methods do not improve, and even slightly lower, the quality of predictions is the Wine quality dataset in Table 47; for this data set the label variable is bounded but with bounds that are far from typical values of the labels.)

Appendix C. Code Availability

Python code for reproducing our experiments in Sect. 8 and Appendix B is available at the GitHub repository <https://github.com/ip200/ivar-experiments> (Petej, 2026).

Table 46: Bias correction dataset

	none	CVAR1	CVAR10
Elastic Net	1.836 ± 0.003	1.590 ± 0.003	1.605 ± 0.003
Gradient Boosting	1.208 ± 0.002	1.272 ± 0.003	1.292 ± 0.003
Lasso	1.977 ± 0.004	1.738 ± 0.003	1.751 ± 0.003
Linear Regression	1.463 ± 0.003	1.507 ± 0.003	1.523 ± 0.003
Random Forest	0.979 ± 0.002	1.078 ± 0.003	1.102 ± 0.003
Ridge	1.463 ± 0.003	1.507 ± 0.003	1.523 ± 0.003
SVR (RBF)	1.152 ± 0.003	1.235 ± 0.003	1.256 ± 0.003
average	1.440	1.418	1.436

Table 47: Wine quality dataset

	none	CVAR1	CVAR10
Elastic Net	0.870 ± 0.001	0.870 ± 0.001	0.870 ± 0.001
Gradient Boosting	0.681 ± 0.001	0.684 ± 0.001	0.684 ± 0.001
Lasso	0.870 ± 0.001	0.870 ± 0.001	0.870 ± 0.001
Linear Regression	0.733 ± 0.001	0.732 ± 0.001	0.732 ± 0.001
Random Forest	0.604 ± 0.002	0.618 ± 0.002	0.619 ± 0.002
Ridge	0.733 ± 0.001	0.732 ± 0.001	0.732 ± 0.001
SVR (RBF)	0.676 ± 0.001	0.677 ± 0.001	0.678 ± 0.001
average	0.738	0.740	0.741

Table 48: Airfoil self-noise dataset

	none	CVAR1	CVAR10
Elastic Net	5.616 ± 0.018	4.865 ± 0.020	4.992 ± 0.019
Gradient Boosting	2.658 ± 0.016	3.146 ± 0.017	3.423 ± 0.017
Lasso	5.541 ± 0.019	5.072 ± 0.022	5.182 ± 0.021
Linear Regression	4.820 ± 0.021	4.670 ± 0.018	4.814 ± 0.018
Random Forest	1.767 ± 0.014	2.626 ± 0.016	2.957 ± 0.017
Ridge	4.820 ± 0.020	4.670 ± 0.018	4.814 ± 0.018
SVR (RBF)	3.797 ± 0.019	4.097 ± 0.018	4.299 ± 0.018
average	4.145	4.164	4.354

Table 49: Student performance dataset

	none	CVAR1	CVAR10
Elastic Net	241.009 ± 0.626	241.617 ± 0.632	241.694 ± 0.640
Gradient Boosting	231.824 ± 0.613	232.643 ± 0.638	232.996 ± 0.646
Lasso	240.262 ± 0.633	240.960 ± 0.641	241.154 ± 0.648
Linear Regression	241.279 ± 0.649	241.408 ± 0.635	241.560 ± 0.643
Random Forest	235.259 ± 0.691	234.641 ± 0.704	234.992 ± 0.702
Ridge	240.821 ± 0.636	241.245 ± 0.634	241.422 ± 0.644
SVR (RBF)	256.059 ± 0.745	245.013 ± 0.653	244.957 ± 0.655
average	240.931	239.647	239.825